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### System and method for validating a digital item token on a simulated user device

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#### Abstract

The disclosure provides a method that includes presenting a simulated user device in a virtual environment, and receiving user credentials in response to an avatar interacting with the simulated user device. The method includes comparing the user credentials to known user credentials. The method includes determining that the avatar is associated with the user based on the comparison. The method includes identifying a first prior interaction from a data log of interactions, where the first prior interaction is associated with a first digital item token, and identifying a second prior interaction from the data log of interactions, where the second prior interaction is associated with a second digital item token. The method includes retrieving the first digital item token and the second digital item token, and presenting the first digital item token and the second digital item token on the simulated user device.

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## Background/Summary

### TECHNICAL FIELD

(1) This disclosure relates generally to network communications and information security. More particularly, this disclosure relates to a system and method for validating a digital item token on a simulated user device.

### BACKGROUND

(2) An organization entity may provide different services at different physical locations within a network to allow users to view a list of items. Existing systems require users to submit credentials at the different physical locations each time to access the services in the network.

### SUMMARY

(3) The system and method described in the present disclosure provide practical applications and technical advantages that overcome the current technical problems described herein. Currently, when a user wants to access a list of digital item tokens, such as non-fungible tokens (NFTs), the user has to log into a real-world user device in the physical world. The provided systems and methods are integrated into the practical application of allowing the user to access a list of digital item tokens on a simulated device within the virtual environment. The provided systems and methods allow for an avatar associated with the user to view a digital item token as soon as it is acquired in the virtual environment. For example, the digital item token can be viewed on a simulated virtual device in the virtual environment itself in real-time, without having to switch to a real-world device to view the same. In some embodiments, the provided system and method include a virtual environment exchange server in communication with the simulated user device in the virtual environment. The virtual environment exchange server is configured to allow the user to view digital item tokens acquired from multiple virtual environments, which may occur in real-time. For example, a digital item token may be acquired in a second virtual environment, and the virtual environment exchange server is configured to allow an avatar associated with the user in the first virtual environment to view digital item tokens on the simulated virtual device as soon as it is acquired in the second virtual environment.

(4) The disclosed system and method provide several practical applications and technical advantages. First, the disclosed system and method lead to the technical advantage of improving an efficiency of application services provided by an entity, and the technical advantage of improving the overall operations of the computing system. For example, the disclosed system and method allow for an avatar of the user to view digital item tokens on a simulated user device in a virtual environment, which may be viewed as soon as the digital item token is acquired in the virtual environment. By allowing the avatar of the user to view the digital item token in real-time within the virtual environment, it avoids having the user switch back to a real-world device to view the same. The disclosed system and method improves the user's access to information within the virtual environment. Second, the virtual-world exchange server improves the user's access to information across multiple virtual environments, thereby increasing the user's access to information within the virtual environment.

(5) In some embodiments, the present disclosure provides a system. The system comprises a memory operable to store a data log of interactions performed by an avatar in a virtual environment, one or more known user credentials associated with a user a first digital item token associated with the virtual environment and a second digital item token associated with the virtual environment. The system comprises a processor operably coupled to the memory. The processor is configured to present a simulated user device in the virtual world, where the simulated user device

is a computer simulation of a real-world user device in the virtual environment. The processor is configured to receive one or more user credentials associated with the user in response to the avatar interacting with the simulated user device in the virtual environment, and compare the one or more user credential associated with the user to the one or more known user credentials in the memory. The processor is configured to authenticate that the avatar is associated with the user based at least in part on the comparison, and in response, the processor is configured to identify a first prior interaction performed by the avatar in the virtual environment from the data log of interactions, where the first prior interaction is associated with the first digital item token. The processor is further configured to identify a second prior interaction performed by the avatar in the virtual environment from the data log of interactions, where the second prior interaction is associated with the second digital item token. The processor is configured to retrieve the first digital item token from the memory, retrieve the second digital item token from the memory; and present the first digital item token and the second digital item token on a display of the simulated user device in the virtual environment.

(6) Certain embodiments of this disclosure may include some, all, or none of these advantages. These advantages and other features will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings and claims.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a more complete understanding of this disclosure, reference is now made to the following brief description, taken in connection with the accompanying drawings and detailed description, wherein like reference numerals represent like parts.

(2) FIG. 1 illustrates a system according to an embodiment of the present disclosure;

(3) FIG. 2 is a block diagram of an example user device of the system of FIG. 1; and

(4) FIG. 3 illustrates a flowchart of one embodiment of a method of operating the system of FIG. 1.

### DETAILED DESCRIPTION

(5) As described above, when a user wants to access a list of digital item tokens, such as non-fungible tokens (NFTs), the user has to log into a real-world user device in the physical world. The provided systems and methods are integrated into the practical application of allowing the user to access a list of digital item tokens on a simulated device within the virtual environment. The provided systems and methods allow for an avatar associated with the user to view a digital item token as soon as it is acquired in the virtual environment. For example, the digital item token can be viewed on a simulated virtual device in the virtual environment itself in real-time, without having to switch to a real-world device to view the same.

(6) System Components

(7) Network

(8) The network **106** may include any interconnecting system capable of transmitting audio, video, signals, data, messages, or any combination of the preceding. The network **106** may include all or a portion of a local area network, a metropolitan area network, a wide area network, an overlay network, a software-defined network a virtual private network, a packet data network (e.g., the Internet), a mobile telephone network (e.g., cellular networks, such as 4G or 5G), a Plain Old Telephone network, a wireless data network (e.g., Wi-Fi, WiGig, WiMax, etc.), a Long Term Evolution network, a Universal Mobile Telecommunications System network, a peer-to-peer network, a Bluetooth<sup>™</sup> network, a Near Field Communication network, a Zigbee<sup>™</sup> network, and/or any other suitable network. The network **106** may be configured to support any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(9) User Device

(10) A user device **102** is a hardware device that is generally configured to provide hardware and software resources to a user. Examples of a user device **102** include, but are not limited to, a virtual reality device, an augmented reality device, a laptop, a computer, a smartphone, a tablet, a smart device, an Internet-of-Things (IoT) device, or any other suitable type of device. The user device **102** may comprise a graphical user interface (e.g., a display), a touchscreen, a touchpad, keys, buttons, a mouse, or any other suitable type of hardware that allows a user to view data and/or to provide inputs into the user device **102**.

(11) Each user device **102** is configured to display a two-dimensional (2D) or three-dimensional (3D) representation of a virtual environment **140** to a user. Each user device **102** is further configured to allow a user to send a request for authenticating the user device **102** to access the virtual environment **140** and to interact with the server **104**.

(12) Each user device **102** is configured to display a two-dimensional (2D) or three-dimensional (3D) representation of a virtual environment **140** to a user. Examples of a virtual environment **140** include, but are not limited to, a graphical or virtual representation of a metaverse, a map, a city, a building interior, a landscape, a fictional location, an alternate reality, or any other suitable type of location or environment. A virtual environment **140** may be configured to use realistic or non-realistic physics for the motion of objects within the virtual environment **140**. Within the virtual environment **140**, each user may be associated with a user device **102** and an avatar **132(a)-(d)**. The avatar **132(a)-(d)** is a graphical representation of the respective user device **102** and the user within the virtual environment **140**. Examples of the avatars **132(a)-(d)** include, but are not limited to, a person, an animal, or an object. In some embodiments, the features and characteristics of the avatar **132(a)-(d)** may be customizable and user defined. For example, the size, shape, color, attire, accessories, or any other suitable type of appearance features may be specified by a user. By using the avatar **132(a)-(d)**, a user or the user device **102** can move within the virtual environment **140** to interact with one or more entities associated with the server **104** or other avatars **132(a)-(d)** and objects within the virtual environment **140**.

(13) FIG. 2 is a block diagram of an embodiment of the user device **102** used by the system **100** of FIG. 1. The user device **102** may be configured to display the virtual environment **140** (referring to FIG. 1) within a field of view of the user (referring to FIG. 1), capture biometric, sensory, and/or physical information of the user wearing and operating the user device **102**, and to facilitate an electronic interaction between the user and the server **104** (referring to FIG. 1).

(14) The user device **102** comprises a processor **202**, a memory **204**, and a display **206**. The processor **202** comprises one or more processors operably coupled to and in signal communication with memory **204**, display **206**, camera **208**, wireless communication interface **210**, network interface **212**, microphone **214**, GPS sensor **216**, and biometric devices **218**. The one or more processors is any electronic circuitry including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate array (FPGAs), application specific integrated circuits (ASICs), or digital signal processors (DSPs). The processor **202** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **202** may be 8-bit, 16-bit, 32-bit, 64-bit or of any other suitable architecture. The processor **202** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations, processor registers that supply operands to the ALU and store the results of ALU operations, and a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers and other components. The one or more processors are configured to implement various instructions. For example, the one or more processors are configured to execute instructions to implement the function disclosed herein, such as some or all of those described with respect to FIGS. 1 and 3. For example, processor **202** may be configured to display virtual objects on display **206**, detect user location, identify virtual sub,

capture biometric information of a user, via one or more of camera **208**, microphone **214**, and/or biometric devices **218**, and communicate via wireless communication interface **210** with server **104** and/or other user devices.

(15) The memory **204** is operable to store any of the information described with respect to FIGS. **1** and **3** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **202**. The memory **204** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. For example, the memory **204** may store the instructions and logic rules **220**, which are described below for performing the functions of user device **102** with respect to FIGS. **1** and **3**.

(16) Display **206** is configured to present visual information to the user (for example, user in FIG. **1**) in an augmented reality environment that overlays virtual or graphical objects onto tangible objects in a real scene in real-time. In other embodiments, the display **206** is configured to present visual information to the user as the virtual environment **140** (referring to FIG. **1**) in real-time. In an embodiment, display **206** is a wearable optical display (e.g., glasses or a headset) configured to reflect projected images and enables a user to see through the display **206**. For example, display **206** may comprise display units, lens, semi-transparent mirrors embedded in an eye glass structure, a visor structure, or a helmet structure. Examples of display units include, but are not limited to, a cathode ray tube (CRT) display, a liquid crystal display (LCD), a liquid crystal on silicon (LCOS) display, a light emitting diode (LED) display, an active matrix OLED (AMOLED), an organic LED (OLED) display, a projector display, or any other suitable type of display as would be appreciated by one of ordinary skill in the art upon viewing this disclosure. In another embodiment, display **206** is a graphical display on a user device **102**. For example, the graphical display may be the display of a tablet or smart phone configured to display an augmented reality environment with virtual or graphical objects overlaid onto tangible objects in a real scene in real-time environment and/or virtual environment **140**.

(17) Camera **208** is configured to capture images of the user wearing the user device **102**. Camera **208** may be configured to capture images continuously, at predetermined intervals, or on-demand. For example, camera **208** may be configured to receive a command from the user to capture an image. In another example, camera **208** is configured to continuously capture images to form a video stream. Camera **208** is communicably coupled to processor **202**.

(18) Examples of wireless communication interface **210** include, but are not limited to, a Bluetooth™ interface, an RFID interface, a near field communication interface, a local area network interface, a personal area network interface, a wide area network interface, a Wi-Fi interface, a Zigbee™ interface, or any other suitable wireless communication interface as would be appreciated by one of ordinary skill in the art upon viewing this disclosure. Wireless communication interface **210** is configured to facilitate processor **202** in communicating with other devices. Wireless communication interface **210** is configured to employ any suitable communication protocol.

(19) The network interface **212** is configured to enable wired and/or wireless communications. The network interface **212** is configured to communicate data between the user device **102** and other network devices, systems, or domain(s). For example, the network interface **212** may comprise a WIFI interface, a local area network interface, a wide area network interface, a modem, a switch, or a router. The processor **202** is configured to send and receive data using the network interface **212**. The network interface **212** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(20) Microphone **214** is configured to capture audio signals (e.g., voice signals or commands) from the user. Microphone **214** is communicably coupled to processor **202**.

(21) GPS sensor **216** is configured to capture and to provide geographical location information. For

example, GPS sensor **216** is configured to provide a geographic location of the user, such as user, employing user device **102**. GPS sensor **216** may be configured to provide the geographic location information as a relative geographic location or an absolute geographic location. GPS sensor **216** may provide the geographic location information using geographic coordinates (i.e., longitude and latitude) or any other suitable coordinate system. GPS sensor **216** is communicably coupled to processor **202**.

(22) Examples of biometric devices **218** may include, but are not limited to, retina scanners and fingerprint scanners. Biometric devices **218** are configured to capture information about a person's physical characteristics and to output a biometric signal based on captured information. Biometric device **218** is communicably coupled to processor **202**.

(23) Server

(24) Referring back to FIG. **1**, the server **104** is a hardware device that is generally configured to provide services and software and/or hardware resources to user devices **102**. The server **104** is generally a server, or any other device configured to process data and communicate with user devices **102** via the network **106**. In particular embodiments, the server **104** may be implemented in the cloud or may be organized in either a centralized or distributed manner.

(25) The server **104** comprises a processor **108** operably coupled with a network interface **110** and a memory **114**. The processor **108** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). For example, one or more processors may be implemented in cloud devices, servers, virtual machines, and the like. The processor **108** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable number and combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **108** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The processor **108** may include an arithmetic logic unit (ALU) for performing arithmetic and logic operations. The processor **108** may register the supply operands to the ALU and store the results of ALU operations. The processor **108** may further include a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers, and other components. The one or more processors are configured to implement various software instructions. In this way, processor **108** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the processor **108** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The processor **108** is configured to operate the server **104** as described in FIGS. **1-3**. For example, the processor **108** may be configured to perform one or more operations of the operational flow **300** as described in FIG. **2**.

(26) The network interface **110** configured to enable wired and/or wireless communications between the server **104** and the network **106**, as well as other components in the system **100**. Suitable network interfaces **110** include an NFC interface, a Bluetooth™ interface, a Zigbee™ interface, a Z-wave interface, a radio-frequency identification (RFID) interface, a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a metropolitan area network (MAN) interface, a personal area network (PAN) interface, a wireless PAN (WPAN) interface, a modem, a switch, and/or a router. The network interface **110** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(27) The memory **114** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The memory **114** may include one or more of a local database, cloud database, network-attached storage (NAS), etc. The memory **114** comprises one or more disks, tape drives, or solid-state drives, and may be used

as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **114** may store any of the information described for the server **104** in FIGS. 1-3 along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **108**.

(28) The memory **114** is operable to store virtual environment information **116** that includes user information **118**, a data log of interactions **120** performed by the user in the virtual environment **140**, one or more known user credentials **122**, a simulated user device **124** having a display **126** for displaying one or more digital item tokens **128**, virtual locations **130(a)-(d)**, avatars **132**, and virtual regions **142(a)-(d)**. The memory **114** is operable to store real world information **134** that includes user information **136** and environment information **138**. The memory **114** is operable to store communication data **144** that includes a user request **146**.

(29) The virtual environment information **116** comprises the user information **118**. The user information **118** generally comprises information that is associated with user accounts that can be used within the virtual environment **140**. For example, the user information **118** may comprise user profile information, online account information, digital assets information, or any other suitable type of information that is associated with the user within a virtual environment **140**. In some embodiments, the user information **118** includes a data log of interactions **120** performed by the avatar **132** of the user in the virtual environment **140**. The data log of interactions **120** may include prior interaction data associated with acquiring one or more digital asset in the virtual environment **140**, such as one or more digital item token **128**. In some embodiments, the digital item token **128** is a non-fungible token (NFT) associated with a digital asset in the virtual environment **140**. The NFT may be associated with any digital asset including, but not limited to, digital currency, digital art, music, film, a virtual item in the virtual environment **140**, virtual real estate in the virtual environment **140**, etc. In one example, the data log of interactions **120** may include data associated with a first prior interaction performed by the avatar **132** in the virtual environment **140** that included acquiring a first digital item token **129**. The data log of interactions **120** may include data associated with a second prior in a second prior interaction performed by the avatar **132** in the virtual environment **140** that included acquiring a second digital item token **131**. Any number of prior interactions and data associated therewith may be stored in the data log of interactions **120**. In some embodiments, the server **104** may update the data log of interactions **120** to include real-time interactions performed by the avatar **132** within the virtual environment **140**. For example, the processor **108** may receive data associated with a third interaction performed by the avatar **132** of the user in the virtual environment **140**, where the third interaction includes acquiring a third digital item token **133**. The processor **108** may update the data log of interactions **120** to store data associated with the third digital item token **133**, which may occur in real-time (e.g., as soon as the avatar **132** acquires the digital asset in the virtual environment **140**). In some embodiments, the user information **118** includes one or more user credentials **122** associated with a user account for a user device to access a virtual environment **140**. The known user credential **122** may comprise login information, such as one or more of user identifiers, username, physical address, email address, phone number, passwords, biometric information associated with the user, and any other data associated with the user account, such as documents, files, media items, etc.

(30) The virtual environment information **116** generally comprises information or data about the appearance of the virtual environment **140**. For example, the virtual environment **140** may comprise information associated with objects, landmarks, buildings, structures, virtual locations **130(a)-(d)**, virtual regions **142(a)-(d)**, or any other suitable type of element that is present within a virtual environment **140**. In some embodiments, the virtual environment information **116** may be used to create a representation of a virtual environment **140** for users. In this case, a virtual environment **140** may be implemented using any suitable type of software framework or engine.

(31) Examples of a virtual environment **140** include, but are not limited to, a graphical or virtual

representation of a metaverse, a map, a city, a building interior, a landscape, a fictional location, an alternate reality, or any other suitable type of location or environment. A virtual environment **140** may be configured to use realistic or non-realistic physics for the motion of objects within the virtual environment **140**. For example, some virtual environment **140** may be configured to use gravity whereas other virtual environment **140** may not be configured to use gravity.

(32) The virtual environment information **116** generally comprises information or data about the appearance of the simulated user device **124**. The simulated user device **124** is a computer simulation of a real-world user device in the virtual environment **140**. The appearance of the simulated user device **124** in the virtual environment **140** may present as any user device including, but not limited to a mobile phone, a smart phone, an electronic tablet device, a computer (e.g., personal computer, desktop, workstation, laptop), and an automated teller machine (ATM). In some embodiments, the simulated user device **124** includes a display **126** configured for displaying digital item tokens **128** received from the data log of interactions **120**. In some embodiments, the simulated user device **124** may be implemented using any suitable type of software framework or engine.

(33) The real-world information **134** comprises user information **136** and environment information **138**. The user information **136** generally comprises information that is associated with user profiles and user accounts that can be used within the real world. For example, user information **136** may comprise user profile information, account information, or any other suitable type of information that is associated with a user within a real-world environment. The user profile comprises one or more known user credentials **122** (e.g., login information) associated with a user account for a user device to access and interact with systems or platforms associated with one or more entities or sub-entities in a physical environment. The environment information **138** generally comprises information that is associated with an entity within the real world that the user is a member of or is associated with. For example, the environment information **138** may comprise physical addresses, GPS based locations, phone numbers, email addresses, contact names, or any other suitable type of information that is associated with one or more entities. Since the server **104** has access to both the virtual environment information **116** and the real-world information **134**, the server **104** may link the virtual environment information **116** and the real-world information **134** together for a user such that changes to the virtual environment information **116** affect or propagate to the real-world information **134** and vice-versa. For example, one or more user login credentials **122** associated with a user account for a user device to access the virtual environment **140** are also used for the user device to access and interact with the systems or platforms associated with one or more entities or sub-entities in a physical environment. The server **104** may be configured to store one or more maps that translate or convert different types of interactions between the real world and the virtual environment **140** and vice-versa.

(34) The server **104** may generate a virtual environment **140** based on the virtual environment information **116** and the real-world information **134**. The virtual environment **140** may be a graphical or virtual representation of a metaverse, a map, a city, a building interior, a landscape, a fictional location, an alternate reality, or any other suitable type of location or environment. As illustrated in FIG. 1, the virtual environment **140** comprises a plurality of virtual locations **130** (e.g., **130a-130d**) associated with corresponding virtual regions **142** (e.g., **142a** and **142b**) and geographical regions. Each virtual location **130** in the virtual environment is associated with at least one entity which operates and provides services in a geographical region in the real-world environment through the network **106**. In one embodiment, each virtual location **130** may be configured and executed by the processor **108** to provide one or more services provided by an entity in a geographical region in the real-world environment. The server **104** may be configured to store one or more maps executed by the processor **108** that translate or convert different types of interactions occurred in the virtual environment **140** between the real world and the virtual environment **140** and vice-versa.

(35) Within the virtual environment **140**, an avatar **132** is generated by the processor **108** as a graphical representation of a user device **102** within the virtual environment **140**. The avatar **132** is associated with corresponding user information **118** which may include one or more user identifiers, username, physical address, passwords, biometric data associated with the user, email address, phone number, documents associated with a user account and an entity. The avatar **132** includes a plurality of features and characteristics which are processed by the processor **108** to present the avatar **132** as the graphical representation of the user device **102** in the virtual environment **140**. For example, the server **104** may receive a signal indicating a physical location of the user device **102** and/or detect the user device **102** in the real-world environment. The server **104** may store the received signal in the memory **114**. The server **104** may determine a virtual location of an avatar **132** associated with the user device **102** in the virtual environment **140** based on the physical location of the user device **102**.

(36) The server **104** may authorize a user device **102** to access the virtual environment **140** to interact with one or more entities at different virtual locations through the server **104**. In an example operation, the server **104** may receive incoming communication data **144** that includes a user request **146** from the avatar **132** through the user device **102** for accessing the virtual environment **140** via the network **106**. The server **104** may receive one or more user credential and authorize the avatar **132** associated with the user device **102** to enter the virtual environment **140**. For example, the server **104** may compare the one or more user credential to known user credentials **122** in the user profile. If they match, the server **104** may authorize the avatar **132** associated with the user device **102** to enter the virtual environment **140**.

(37) Virtual Environment Exchange Server

(38) In some embodiments, the system **100** includes a virtual environment exchange server **148** in communication with the server **104** via the network **106**. The virtual environment exchange server **148** is configured to communicate data and information between the server **104** and a second virtual environment server **205** that hosts a second virtual environment **240**. The virtual environment exchange server **148** may communicate data and information between the server **104** and any number of virtual environment servers (e.g., up to an N.sup.th virtual environment server **304** that hosts an N.sup.th virtual environment **340**). In general, the virtual environment exchange server **148** allows the avatar **132** of the user in the virtual environment **140** to view digital item tokens **228**, **328** that were acquired by avatars **232**, **332** associated with the user on different virtual environments hosted by different virtual environment servers (e.g., a second virtual environment **240** hosted by a second virtual environment server **205** to an N.sup.th virtual environment **340** hosted by an N.sup.th virtual environment server **304**). The virtual-environment exchange server **148** may update the data log of interactions **120** on the server **104** to include digital item tokens **228** from the second virtual environment **240** and/or the digital item tokens **328** from the N.sup.th virtual environment **340**. This advantageously allows the avatar **132** of the user in the virtual environment **140** to view assets the user has acquired across multiple virtual environments within a single virtual environment, and avoids having the user login to the other virtual environments to check on the digital assets.

(39) The virtual environment exchange server **148** includes a processor **150** operably coupled with a network interface **152** and a memory **154**. The processor **150** is any electronic circuitry, including, but not limited to, state machines, one or more central processing unit (CPU) chips, logic units, cores (e.g., a multi-core processor), field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), or digital signal processors (DSPs). For example, one or more processors may be implemented in cloud devices, servers, virtual machines, and the like. The processor **150** may be a programmable logic device, a microcontroller, a microprocessor, or any suitable number and combination of the preceding. The one or more processors are configured to process data and may be implemented in hardware or software. For example, the processor **150** may be 8-bit, 16-bit, 32-bit, 64-bit, or of any other suitable architecture. The processor **150** may

include an arithmetic logic unit (ALU) for performing arithmetic and logic operations. The processor **150** may register the supply operands to the ALU and store the results of ALU operations. The processor **150** may further include a control unit that fetches instructions from memory and executes them by directing the coordinated operations of the ALU, registers, and other components. The one or more processors are configured to implement various software instructions. In this way, processor **150** may be a special-purpose computer designed to implement the functions disclosed herein. In an embodiment, the processor **150** is implemented using logic units, FPGAs, ASICs, DSPs, or any other suitable hardware. The processor **150** is configured to operate as described in FIGS. **1-3**. For example, the processor **150** may be configured to perform one or more operations of the operational flow **300** as described in FIG. **3**.

(40) The network interface **152** configured to enable wired and/or wireless communications between the virtual environment exchange server **148**, the second virtual environment server **205**, the N.sup.th virtual environment server **304** and the network **106**. Suitable network interfaces **152** include an NFC interface, a Bluetooth™ interface, a Zigbee™ interface, a Z-wave interface, a radio-frequency identification (RFID) interface, a WIFI interface, a local area network (LAN) interface, a wide area network (WAN) interface, a metropolitan area network (MAN) interface, a personal area network (PAN) interface, a wireless PAN (WPAN) interface, a modem, a switch, and/or a router. The network interface **152** may be configured to use any suitable type of communication protocol as would be appreciated by one of ordinary skill in the art.

(41) The memory **154** may be volatile or non-volatile and may comprise read-only memory (ROM), random-access memory (RAM), ternary content-addressable memory (TCAM), dynamic random-access memory (DRAM), and static random-access memory (SRAM). The memory **154** may include one or more of a local database, cloud database, network-attached storage (NAS), etc. The memory **154** comprises one or more disks, tape drives, or solid-state drives, and may be used as an over-flow data storage device, to store programs when such programs are selected for execution, and to store instructions and data that are read during program execution. The memory **154** may store any of the information described in FIGS. **1-3** along with any other data, instructions, logic, rules, or code operable to implement the function(s) described herein when executed by processor **150**.

(42) The memory may store user information **318** that is associated with user accounts that can be used within the second virtual environment **240** and the N.sup.th virtual environment **340**. For example, the user information **318** may comprise user profile information, online account information, digital assets information, or any other suitable type of information that is associated with the user within the second virtual environment **240** and the N.sup.th virtual environment **340**. In some embodiments, the user information **318** includes a data log of interactions **320** performed by the avatars **232**, **323** of the user within the second virtual environment **240** and the N.sup.th virtual environment **340**. The data log of interactions **320** may include prior interaction data associated with acquiring one or more digital asset in the second virtual environment **240** and the N.sup.th virtual environment **340**, such as one or more digital item token **228**, **328**. In some embodiments, the virtual environment exchange server **148** may update the data log of interactions **120** of the virtual environment **140** to include one or more digital item token **228**, **328**. In some embodiments, the user information **318** includes one or more known user credentials **122** associated with a user account for a user device **102** to access in the second virtual environment **240** and the N.sup.th virtual environment **340**. The known user credential **122** may comprise login information, such as one or more of user identifiers, username, physical address, email address, phone number, passwords, biometric information associated with the user, and any other data associated with the user account, such as documents, files, media items, etc.

(43) System Operation

(44) FIG. **3** illustrates one embodiment of a flowchart **400** for presenting one or more digital item tokens **128** to an avatar **132** of a user on a simulated user device **124** in a virtual environment **140**.

The operational flow **400** can be logically described in two parts. The first part includes presenting a simulated user device **124** in a virtual environment **140**, receiving one or more user credential in response to an avatar **132** interacting with the simulated user device **124** in the virtual environment **140**, and comparing the one or more user credential to one or more known user credential **122** in the memory **114** of the server **104** to determine whether the avatar **132** is associated with the user based at least in part on the comparison. If the user is authorized, the operational flow proceeds to the second part. The second part includes identifying a first prior interaction performed by the avatar **132** in the virtual environment **140** from a data log of interactions **120**, identifying a second prior interaction performed by the avatar in the virtual environment **140** from the data log of interactions **120**, retrieving a first digital item token **129** associated with the first prior interaction from the data log of interactions **120**, retrieving a second digital item token **131** associated with the second prior interaction from the data log of interactions **120**, and presenting the first digital item token **129** and the second digital item token **131** on the display **126** of the simulated user device **124** in the virtual environment **140**.

(45) In operation, the flow **400** may begin at operation **402** where the processor **108** of the server **104** is configured to present the simulated user device **124** in the virtual environment **140**. The simulated user device **124** is a computer simulation of a real-world user device in the virtual environment **140**. The appearance of the simulated user device **124** in the virtual environment **140** may present as any user device including, but not limited to a mobile phone, a smart phone, an electronic tablet device, a computer (e.g., personal computer, desktop, workstation, laptop), and an automated teller machine (ATM). In some embodiments, the simulated user device **124** may be presented in the virtual environment **140** by the processor **108** using any suitable type of software framework or engine.

(46) At operation **404**, the processor **108** receives one or more user credential in response to the avatar **132** interaction with the simulated user device **124** in the virtual environment **140**. For example, the avatar **132** of the user may approach the simulated user device **124** in the virtual environment **140** and enter one or more user credential, such as a username and password, in attempts to login to the simulated user device **124**. At operation **406**, the processor **108** compares the one or more user credentials provided by the avatar **132** to the known user credentials **122** stored in the memory **114**. At decision block **408**, the processor **108** may determine whether the avatar **132** authorized to access the simulated user device **124** based on the comparison in operation **406**. If the one or more user credential provided by the avatar **132** do not match the known user credentials **122**, then operational flow **400** returns to operation **404** where the avatar **132** is prompted to re-enter the one or more user credentials. If the one ore more user credentials provided by the avatar **132** match the known user credentials **12**, then operational flow **400** proceeds to operation **410**.

(47) At operation **410**, the processor **108** identifies a first prior interaction performed by the avatar **132** in the virtual environment **140** that includes a first digital item token **129**, and the processor **108** further identifies a second prior interaction performed by the avatar **132** in the virtual environment **140** that includes a second digital item token **131**. The prior interactions may be identified from the data log of interactions **120** by the processor **108**. In some embodiments, the first prior interaction and second prior interaction performed by the avatar **132** in the virtual environment **140** may include a previous action where the avatar **132** acquired (e.g., purchased) the first digital item token **129** and the second digital item token **131** in the virtual environment **140**. In some embodiments, the digital item token **128** is a non-fungible token (NFT) associated with a digital asset in the virtual environment **140**. The NFT may be associated with any digital asset including, but not limited to, digital currency, digital art, music, film, a virtual item in the virtual environment **140**, and virtual real estate in the virtual environment **140**.

(48) At operation **412**, the processor **108** retrieves the first digital item token **129** associated with the first prior interaction and further retrieves the second digital item token **131** associated with the

second prior interaction from the data log of interactions **120**. The processor **108** is further configured to communicate the first digital item token **129** and the second digital item token **131** to the simulated user device **124**. At operation **414**, the processor **108** is configured to present the first digital item token **129** and the second digital item token **131** on the display **126** of the simulated user device **124** in the virtual environment **140** to the avatar **132** of the user. In one exemplary use case, the simulated user device **124** may be an automated teller machine that is configured to display the one or more digital item tokens **128** associated with the user within the virtual environment **140**.

(49) In some embodiments, the processor **108** may update the data log of interactions **120** to include real-time interactions performed by the avatar **132** within the virtual environment **140**. For example, the processor **108** may receive data associated with a third interaction performed by the avatar **132** of the user in the virtual environment **140**, where the third interaction includes acquiring a third digital item token **133**. The processor **108** may update the data log of interactions **120** to store data associated with the third digital item token **133**, which may occur in real-time (e.g., as soon as the avatar **132** acquires the digital asset in the virtual environment **140**). In some embodiments, operation **410** may further include identifying a third interaction in the data log of interactions **120**. The third interaction may be updated in the data log of interactions **120** in real time (e.g., occurred during a current session while the avatar **132** is active in the virtual environment **140**). In other words, the third interaction may appear in the data log of interactions **120** without having the user log out of the virtual environment **140** and relog back in as the avatar **132**. Operation **412** may further include having the processor **108** retrieve the third digital item token **133** from the data log of interactions **120**, and operation **414** may further include presenting the third digital item token **133** on a display **126** of the simulated user device **124** in the virtual environment.

(50) In some embodiments, at operation **410**, the processor **108** is further configured to identify a prior interaction performed by a second avatar **232** associated with the user in the second virtual environment **240**, where the prior interaction is associated with a first digital item token **229** from the second virtual environment **240**. In some embodiments, the prior interaction from the second virtual environment and the associated first digital item token **229** may be stored in a second data log of interactions **320** a second memory **154** of the virtual environment exchange server **148**. At operation **412**, the processor **108** is further configured to retrieve the digital item token **228** associated with the second virtual environment **240** from the second memory **154** of the virtual environment exchange server **148**, and communicate the digital item token **228** to the simulated user device **124**. In some embodiments, operation **412** includes updating the data log of interactions **120** in the virtual environment **140** to include the digital item token **228** from the second virtual environment **240**. In some embodiments, operation **414** further includes presenting the digital item token **228** on the display **126** of the simulated user device **124** in the virtual environment **140**.

(51) In some embodiments, the processor **150** of the virtual environment exchange server **148** may receive data associated with a second interaction performed by the second avatar **232** performed in the second virtual environment **240**, where the second interaction includes a second digital item token **231**. The processor **150** of the virtual environment exchange server **148** may communicate the second interaction including the second digital item token **231** to the server **104**, and update the data log of interactions **120** to include the second digital item token **231** from the second virtual environment **240**. The processor **108** may then retrieve the second digital item token **231** of the second virtual environment from the memory **114** and present the second digital item token **231** of the second virtual environment on the display **126** of the simulated user device **124** in the virtual environment **140**.

(52) While several embodiments have been provided in the present disclosure, it should be understood that the disclosed systems and methods might be embodied in many other specific forms without departing from the spirit or scope of the present disclosure. The present examples are

to be considered as illustrative and not restrictive, and the intention is not to be limited to the details given herein. For example, the various elements or components may be combined or integrated with another system or certain features may be omitted, or not implemented. In addition, techniques, systems, subsystems, and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules, techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as coupled or directly coupled or communicating with each other may be indirectly coupled or communicating through some interface, device, or intermediate component whether electrically, mechanically, or otherwise. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed herein.

(53) To aid the Patent Office, and any readers of any patent issued on this application in interpreting the claims appended hereto, applicants note that they do not intend any of the appended claims to invoke 35 U.S.C. § 112(f) as it exists on the date of filing hereof unless the words “means for” or “step for” are explicitly used in the particular claim.

## Claims

1. A system comprising: a memory operable to store a data log of interactions performed by an avatar in a virtual environment, one or more user credentials associated with a user, a first digital item token associated with the virtual environment, and a second digital item token associated with the virtual environment; and a processor operably coupled to the memory and configured to: present a simulated user device in the virtual environment, wherein the simulated user device is a computer simulation of a real-world user device in the virtual environment; receive one or more user credentials in response to the avatar interacting with the simulated user device in the virtual environment; compare the one or more user credentials to one or more known user credentials stored in the memory; determine that the avatar is associated with the user based at least in part on the comparison, and in response: activate a first communication session between the avatar and the virtual environment; identify a first prior interaction performed by the avatar in the virtual environment from the data log of interactions, wherein the first prior interaction is associated with the first digital item token; identify a second prior interaction performed by the avatar in the virtual environment from the data log of interactions, wherein the second prior interaction is associated with the second digital item token; retrieve the first digital item token from the memory; retrieve the second digital item token from the memory; and present the first digital item token and the second digital item token on display of the simulated user device in the virtual environment; a virtual environment exchange server, comprising: a second memory operable to store a second data log of interactions performed by a second avatar associated with the user in a second virtual environment, the second data log of interactions including a third digital item token associated with the second virtual environment; and a second processor configured to: activate a second communication session between the second avatar and the second virtual environment; update the data log of interactions based on the second data log of interactions, wherein the data log of interactions is updated to include the third digital item token associated with the second virtual environment while both the first communication session and the second communication session continue to remain active; retrieve the first digital item token, the second digital item token, and the third digital item token from the updated data log of interactions while the second communication session is active; and present, over the first communication session, the first digital item token, the second digital item token, and the third digital item token on the display of the simulated user device in the virtual environment while the first communication session and the second communication session continue to remain active; a second virtual environment server, comprising: a third memory operable to store a third data log of interactions performed by a third avatar

associated with the user in a third virtual environment, the third data log of interactions including a fourth digital item token associated with the third virtual environment; and a third processor configured to: activate a third communication session between the third avatar and the third virtual environment; re-update the data log of interactions based on the third data log of interactions, wherein the data log of interactions is re-updated to include the fourth digital item token associated with the third virtual environment while both the first communication session, the second communication session, and third communication session continue to remain active; the second processor of the virtual environment exchange server is further configured to: retrieve the third digital item token and the fourth digital item token from the re-updated data log of interactions over the active first communication session; and present, over the active first communication session, the third digital item token and the fourth digital item token on the display of the simulated user device in the virtual environment while the second communication session and the third communication session continue to remain active.

2. The system of claim 1, wherein each of the first digital item token and the second digital item token comprises a non-fungible token associated with a digital asset in the virtual environment.

3. The system of claim 1, wherein the simulated user device is an automated teller machine in the virtual environment.

4. The system of claim 1, wherein the one or more user credentials comprise one or more of a username, a password, or biometric information associated with the user.

5. The system of claim 1, wherein the virtual environment is a first metaverse, the second virtual environment is a second metaverse, and the third virtual environment is a third metaverse.

6. The system of claim 1 further comprising: identify a prior interaction performed by the second avatar in the second virtual environment from the second data log of interactions, wherein the prior interaction is associated with the third digital item token associated with the second virtual environment; retrieve the third digital item token associated with the second virtual environment from the second memory; and present the third digital item token associated with the second virtual environment on the display of the simulated user device in the second virtual environment.

7. The system of claim 6, wherein the processor is configured to: receive data associated with a second interaction performed by the second avatar in the second virtual environment, wherein the second interaction is associated with a fifth digital item token of the second virtual environment; store the second interaction performed by the second avatar in the data log of interactions in the memory; retrieve the fifth digital item token of the second virtual environment from the memory; and present the fifth digital item token of the second virtual environment on the display of the simulated user device in the virtual environment.

8. A method comprising: presenting a simulated user device in a virtual environment, wherein the simulated user device is a computer simulation of a real-world user device in the virtual environment; receiving one or more user credentials associated with a user in response to an avatar interacting with the simulated user device in the virtual environment; comparing the one or more user credentials associated with the user to one or more known user credentials stored in a memory; determining that the avatar is associated with the user based at least in part on the comparison, and in response: activating a first communication session between the avatar and the virtual environment; identifying a first prior interaction performed by the avatar in the virtual environment from a data log of interactions stored in the memory, wherein the first prior interaction is associated with a first digital item token; identifying a second prior interaction performed by the avatar in the virtual environment from the data log of interactions, wherein the second prior interaction is associated with a second digital item token; retrieving the first digital item token from the memory; retrieving the second digital item token from the memory; and presenting the first digital item token and the second digital item token on a display of the simulated user device in the virtual environment; receiving, from a virtual environment exchange server, a second data log of interactions performed by a second avatar associated with the user in a second virtual environment,

the second data log of interactions including a third digital item token associated with the second virtual environment; activating, by the virtual environment exchange server, a second communication session between the second avatar and the second virtual environment; updating, by the virtual environment exchange server, the data log of interactions based on the second data log of interactions, wherein the data log of interactions is updated to include the third digital item token associated with the second virtual environment while both the first communication session and the second communication session continue to remain active; retrieving the first digital item token, the second digital item token, and the third digital item token from the updated data log of interactions while the second communication session is active; and presenting, over the first communication session, the first digital item token, the second digital item token, and the third digital item token on the display of the simulated user device in the virtual environment while the first communication session and the second communication session continue to remain active; receiving, from a second virtual environment server, a third data log of interactions performed by a third avatar associated with the user in a third virtual environment, the third data log of interactions including a fourth digital item token associated with the third virtual environment; activating, by the second virtual environment server, a third communication session between the third avatar and the third virtual environment; re-updating, by the second virtual environment server, the data log of interactions based on the third data log of interactions, wherein the data log of interactions is re-updated to include the fourth digital item token associated with the third virtual environment while both the first communication session, the second communication session, and third communication session continue to remain active; retrieving, by the virtual environment exchange server, the third digital item token and the fourth digital item token from the re-updated data log of interactions over the active first communication session; and presenting, by the virtual environment exchange server, over the active first communication session, the third digital item token and the fourth digital item token on the display of the simulated user device in the virtual environment while the second communication session and the third communication session continue to remain active.

9. The method of claim 8, wherein each of the first digital item token and the second digital item token comprises a non-fungible token associated with a digital asset in the virtual environment.

10. The method of claim 8, wherein the simulated user device is an automated teller machine in the virtual environment.

11. The method of claim 8, wherein the one or more user credentials comprises one or more of a username, a password, or biometric information associated with the user.

12. The method of claim 8 wherein the virtual environment is a first metaverse, the second virtual environment is a second metaverse, and the third virtual environment is a third metaverse.

13. The method of claim 8 further comprising: retrieving the third digital item token associated with the second virtual environment from the virtual environment exchange server; and presenting the third digital item token associated with the second virtual environment on the display of the simulated user device in the second virtual environment.

14. The method of claim 13 further comprising: receiving, using the virtual environment exchange server, data associated with a second interaction performed by the second avatar in the second virtual environment, wherein the second interaction is associated with a fifth digital item token of the second virtual environment; storing the second interaction performed by the second avatar in the data log of interactions in the memory; retrieving the fifth digital item token of the second virtual environment from the memory; and presenting the fifth digital item token of the second virtual environment on the display of the simulated user device in the virtual environment.

15. A non-transitory computer-readable medium storing instructions that when executed by a processor cause the processor to: present a simulated user device in a virtual environment, wherein the simulated user device is a computer simulation of a real-world user device in the virtual environment; receive one or more user credentials associated with a user in response to an avatar interacting with the simulated user device in the virtual environment; compare the one or more user

credentials associated with the user to one or more known user credentials stored in a memory; determine that the avatar is associated with the user based at least in part on the comparison, and in response: activate a first communication session between the avatar and the virtual environment; identify a first prior interaction performed by the avatar in the virtual environment from a data log of interactions stored in the memory, wherein the first prior interaction is associated with a first digital item token; identify a second prior interaction performed by the avatar in the virtual environment from the data log of interactions, wherein the second prior interaction is associated with a second digital item token; retrieve the first digital item token from the memory; retrieve the second digital item token from the memory; and present the first digital item token and the second digital item token on a display of the simulated user device in the virtual environment; a second processor configured to: receive a second data log of interactions performed by a second avatar associated with the user in a second virtual environment, the second data log of interactions including a third digital item token associated with the second virtual environment; activate a second communication session between the second avatar and the second virtual environment; update the data log of interactions based on the second data log of interactions, wherein the data log of interactions is updated to include the third digital item token associated with the second virtual environment while both the first communication session and the second communication session continue to remain active; retrieve the first digital item token, the second digital item token, and the third digital item token from the updated data log of interactions while the second communication session is active; and present, over the first communication session, the first digital item token, the second digital item token, and the third digital item token on the display of the simulated user device in the virtual environment while the first communication session and the second communication session continue to remain active; receive a third data log of interactions performed by a third avatar associated with the user in a third virtual environment, the third data log of interactions including a fourth digital item token associated with the third virtual environment; activate a third communication session between the third avatar and the third virtual environment; re-update the data log of interactions based on the third data log of interactions, wherein the data log of interactions is re-updated to include the fourth digital item token associated with the third virtual environment while both the first communication session, the second communication session, and third communication session continue to remain active; retrieve the third digital item token and the fourth digital item token from the re-updated data log of interactions over the active first communication session; and present, over the active first communication session, the third digital item token and the fourth digital item token on the display of the simulated user device in the virtual environment while the second communication session and the third communication session continue to remain active.

16. The non-transitory computer-readable medium of claim 15, wherein each of the first digital item token and the second digital item token comprises a non-fungible token associated with a digital asset in the virtual environment.

17. The non-transitory computer-readable medium of claim 15, wherein the simulated user device is an automated teller machine in the virtual environment.

18. The non-transitory computer-readable medium of claim 15, wherein the one or more user credentials comprises one or more of a username, a password, or biometric information associated with the user.

19. The non-transitory computer-readable medium of claim 15, wherein the virtual environment is a first metaverse, the second virtual environment is a second metaverse, and the third virtual environment is a third metaverse.

20. The non-transitory computer-readable medium of claim 19, wherein the instructions further cause the processor to: retrieve the third digital item token associated with the second virtual environment from the virtual environment exchange server; and present the third digital item token

associated from the second virtual environment on the display of the simulated user device in the second virtual environment.

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